

Can BCA identify harmful unfamiliar actions?

Offline agents cannot try new actions during training. If they overestimate an unfamiliar action, they can learn to prefer a poor decision.

BCA uses prediction-error bands to adjust the host algorithm's training penalty. We are testing whether those bands provide useful warnings about harmful actions.

We first compare each host with standard BCA, using equal calibration-fitting weights. Weighted conformal prediction is the next separate method to validate.

The goal is reliable uncertainty that helps the agent make better decisions.

The OOD test compares actions at the same state

We restore a saved simulator state and try the host action, the BCA action, and other candidate actions. We keep the later policy fixed within each comparison.

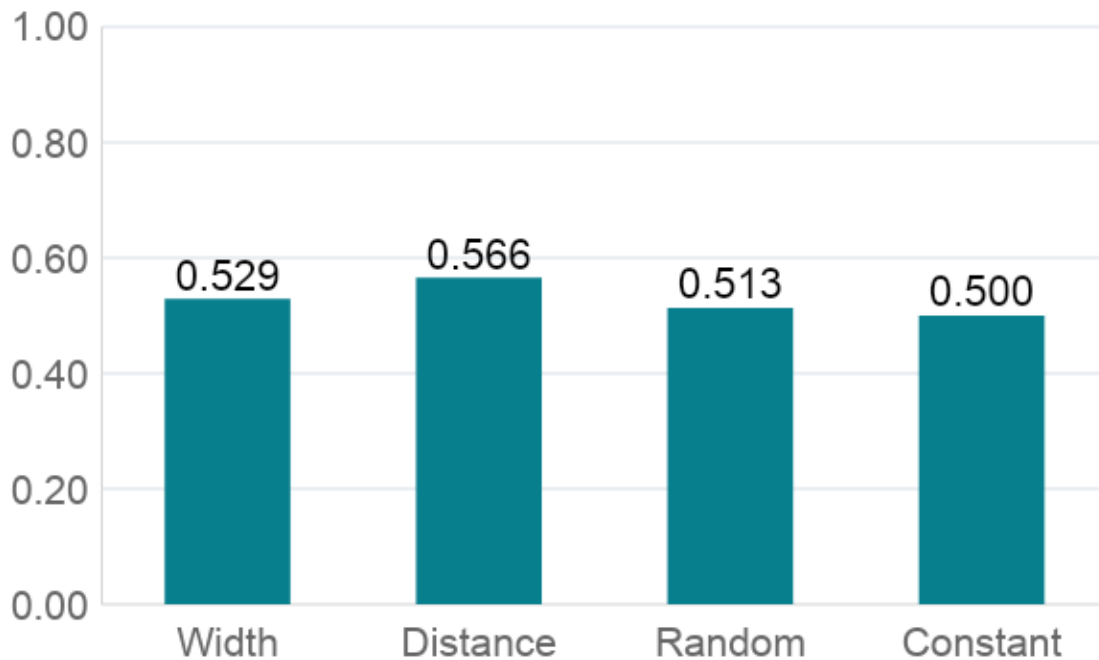
We call an alternative harmful when its continuation earns more than 1 raw reward unit less than the host action. Larger BCA bands should rank those alternatives as riskier.

We also measure distance from recorded actions and regret relative to the best tested action. These distinguish unfamiliarity, warning quality, and decision quality.

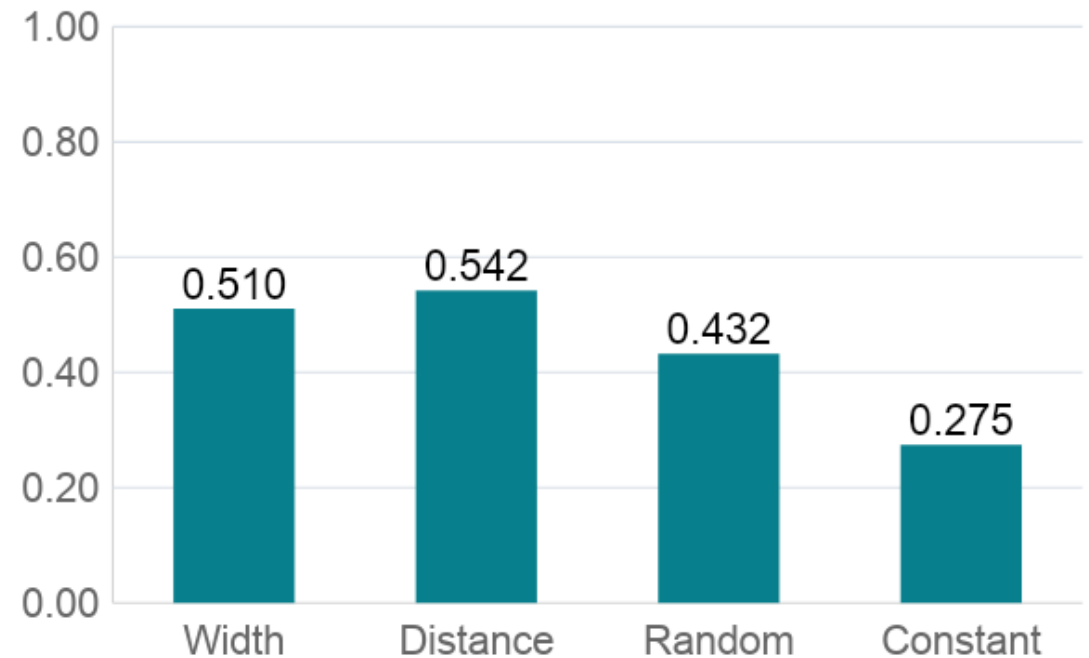
Calibration uses recorded transitions; this separate simulator study measures action outcomes.

BCA's harm warnings are weak in this pilot

AUROC: does a harmful action get a higher warning?



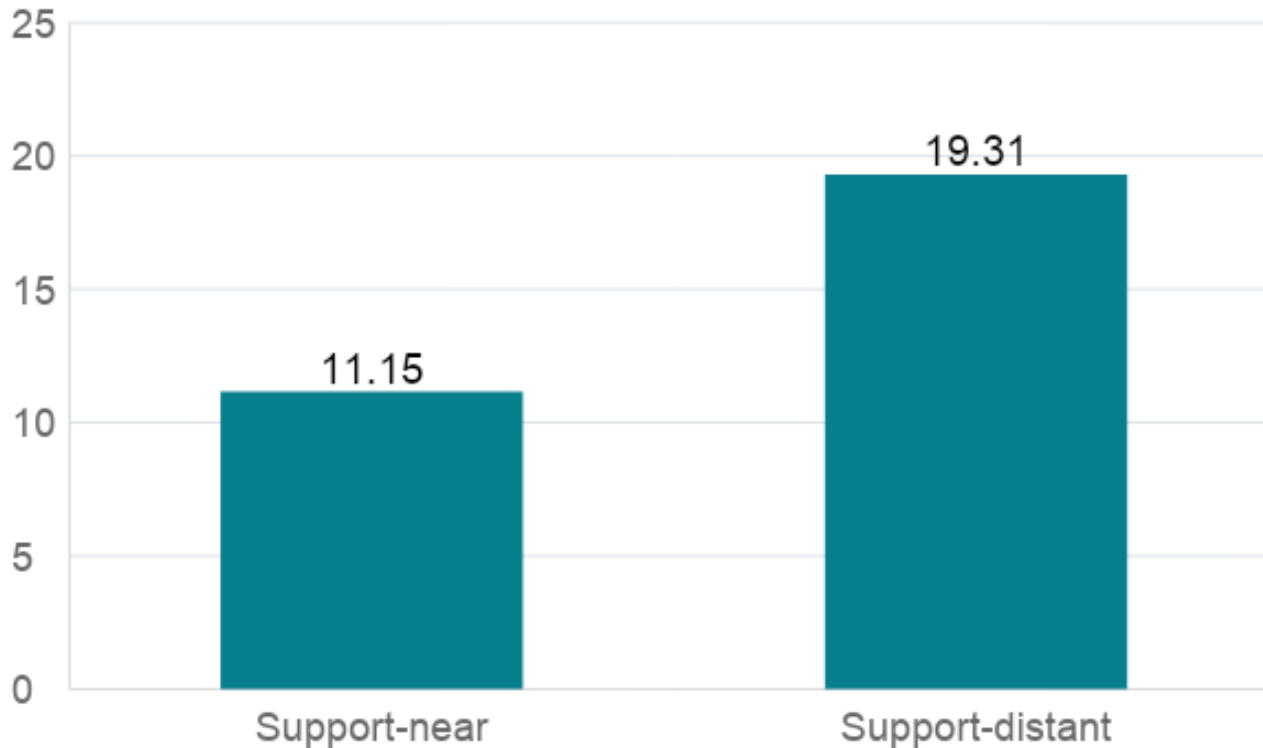
Average precision: are the top warnings useful?



Width AUROC is 0.529; chance is 0.500. Distance scores 0.566.

Unfamiliarity raises risk, but does not define harm

Observed harmful-action rate (%)



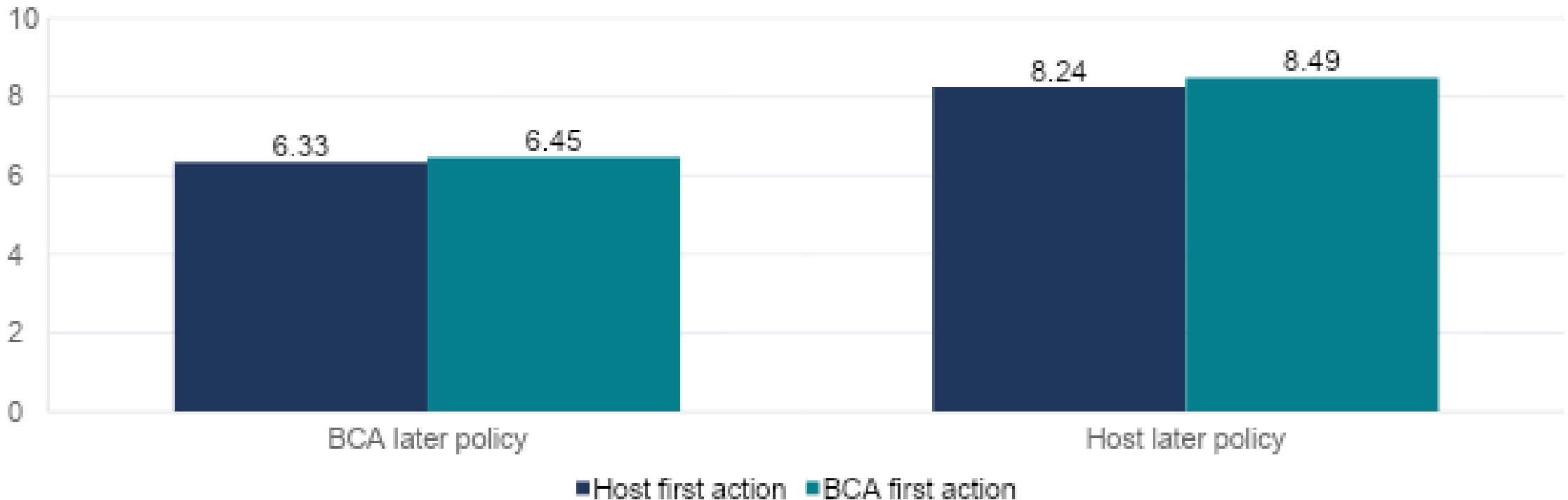
Actions farther from recorded behavior were harmful more often: 19.3% versus 11.2%.

Yet 231 of 276 harmful actions were near recorded behavior. A useful warning must also distinguish harmful from helpful actions within each group.

Distance helps describe the test; simulator outcomes establish whether an action is harmful.

BCA has not improved tested-action regret

Regret = best tested return – selected-action return; lower is better

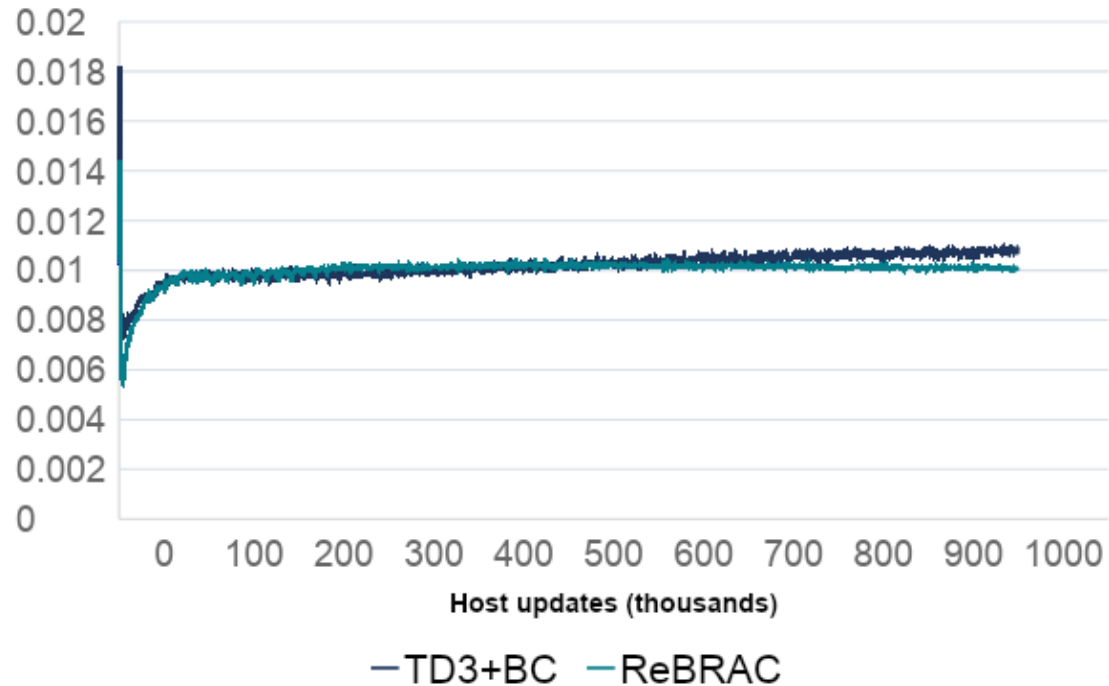


With BCA continuing afterward, mean regret is 6.33 for host actions and 6.45 for BCA actions.

BCA is fitting and changing the training penalty

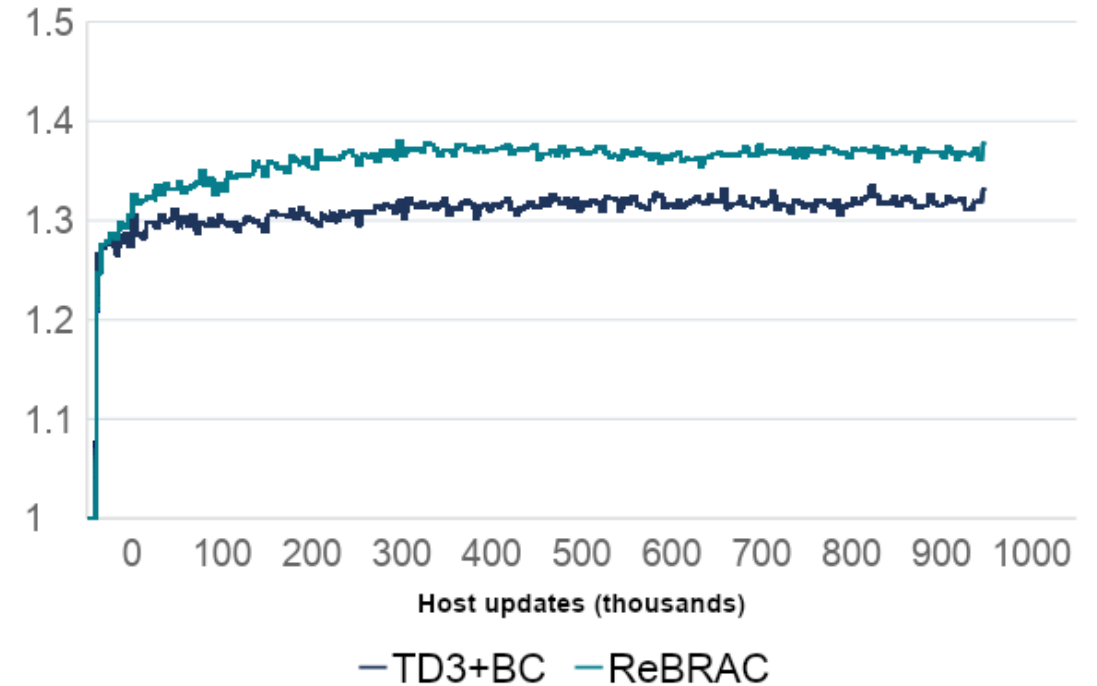
Scale-fitting objective

Block mean objective



Actor behavior-cloning multiplier

Multiplier; 1 = unchanged strength

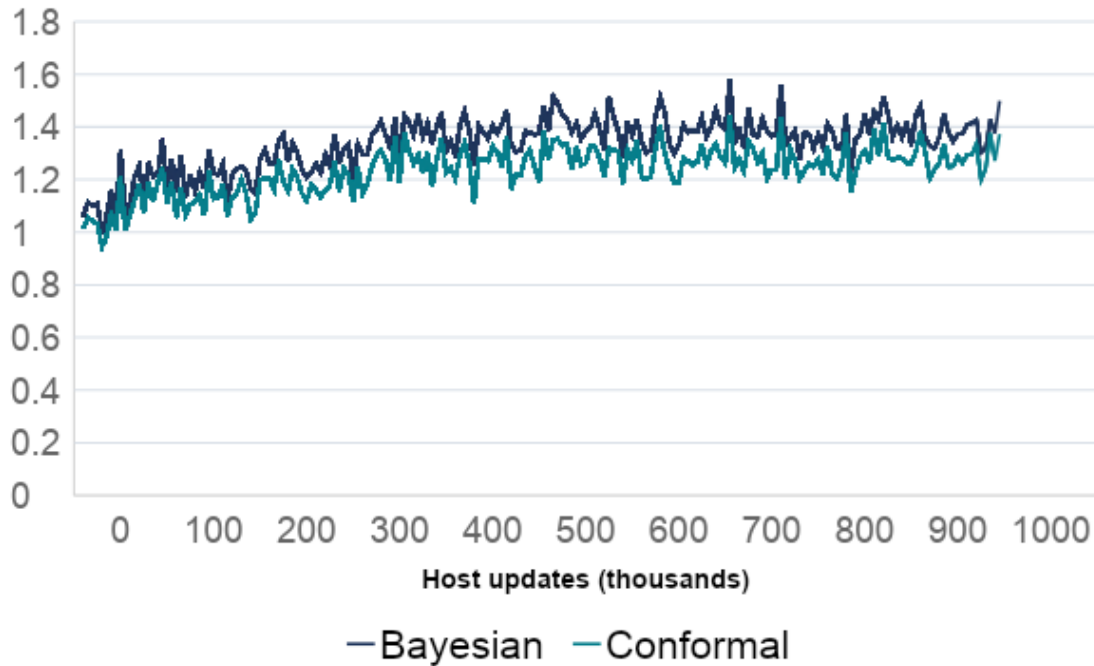


Both runs accepted 1M scale fits. Final BC strength rises by 33.1% and 37.7%.

The Bayesian radius controls these bands

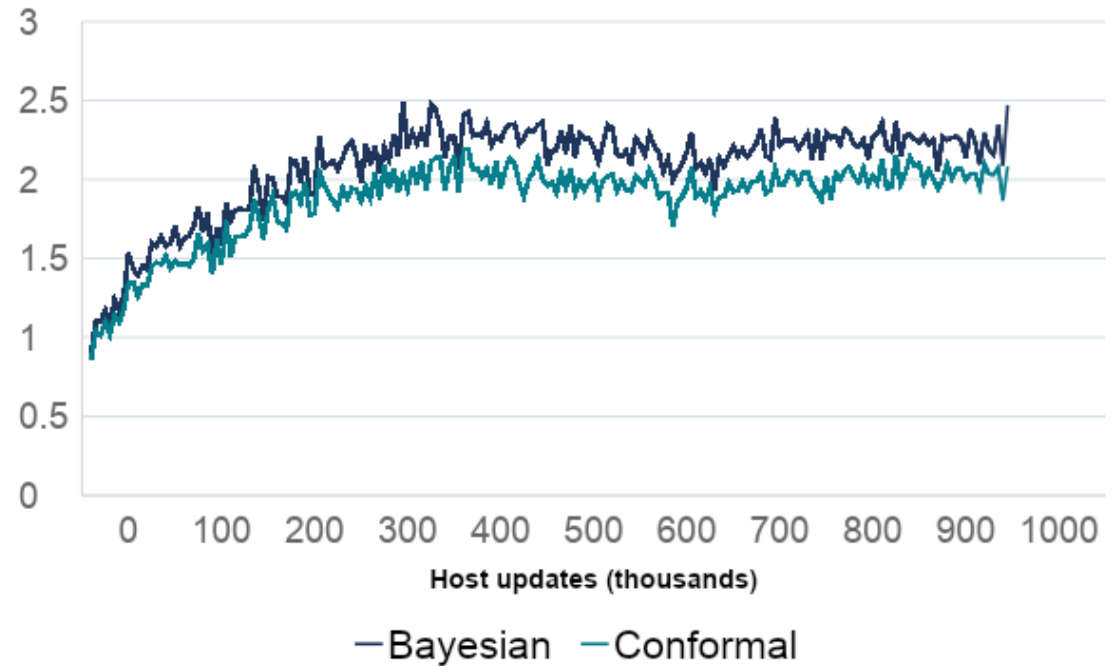
TD3+BC · Walker

Radius in normalized residual units



ReBRAC · Walker

Radius in normalized residual units



The Bayesian radius exceeds the conformal radius at all 198 refreshes in each run.

A corrected conformal threshold changes the final band only when it exceeds the Bayesian radius.

Reviewed TD3+BC/ReBRAC Walker runs. Final radius = $\max(\text{Bayesian radius}, \text{conformal radius})$.

The next question is calibration under shift

The pilot gives weak evidence of useful harm ranking and no mean regret improvement. It does not yet tell us whether the bands achieve their stated coverage.

An offline calibration set may emphasize different state–action situations from those the policy queries. A threshold fitted to the old mixture can miss too many new targets.

Weighted conformal prediction tests that specific distribution-shift problem. We still need separate evidence that correcting coverage improves action choices.

This is a testable explanation to investigate, not an established cause of the pilot result.

Our earlier IW path did not complete weighted CP

The earlier IW variants changed which examples trained the error-scale network. The host path still gave calibration errors equal mass when calculating the conformal threshold.

Weighted CP must also weight those held-out errors by target probability divided by source probability. It must include the queried state–action pair’s weight in the cutoff.

We need justified ratios, a query-dependent threshold, and fresh calibration data after freezing the model. Existing affinity or policy-density weights do not establish all three.

We will identify the corrected weighted-threshold method separately from prior fitting-IW runs.

A controlled test isolates the effect of the shift

We created easy cases with errors between 0 and 1, and hard cases with errors between 0 and 6. Calibration sees 80% easy cases; testing sees only 20%.

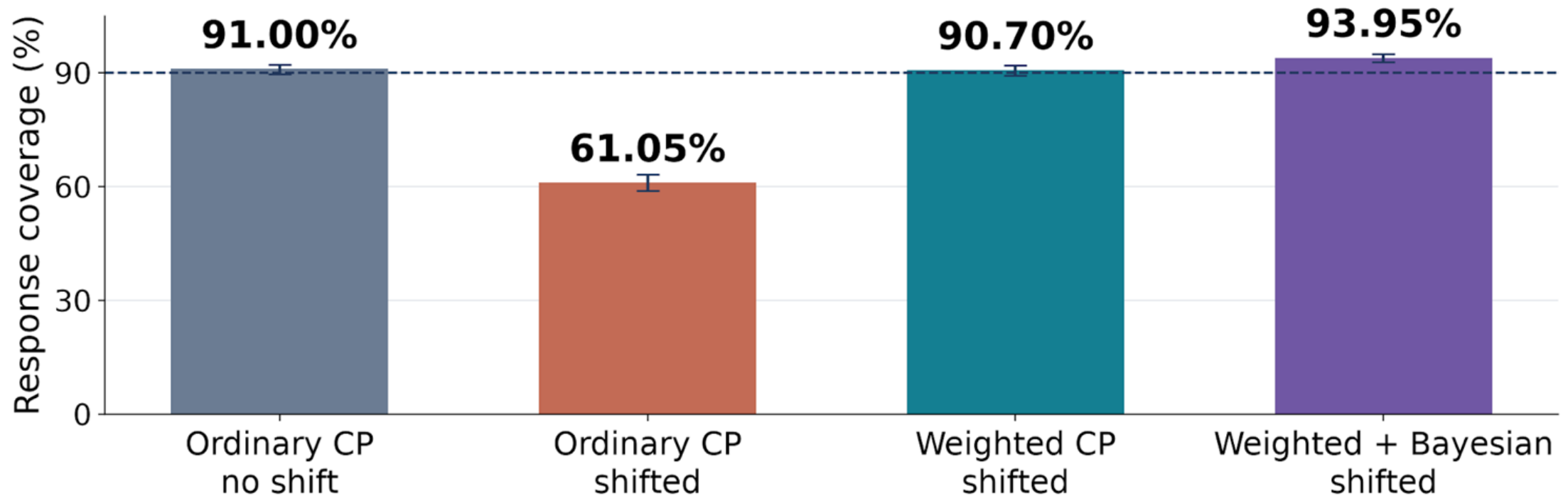
Everything within each case type stays the same. Only the mixture changes, so the correct importance weights are known: easy = $0.2/0.8 = 0.25$; hard = $0.8/0.2 = 4$.

We compare ordinary CP, weighted CP, and weighted CP with the Bayesian component. Each method targets 90% coverage using the same simulated examples.

This isolates the calibration calculation before introducing RL training or estimated ratios.

Weighting restores coverage in the synthetic test

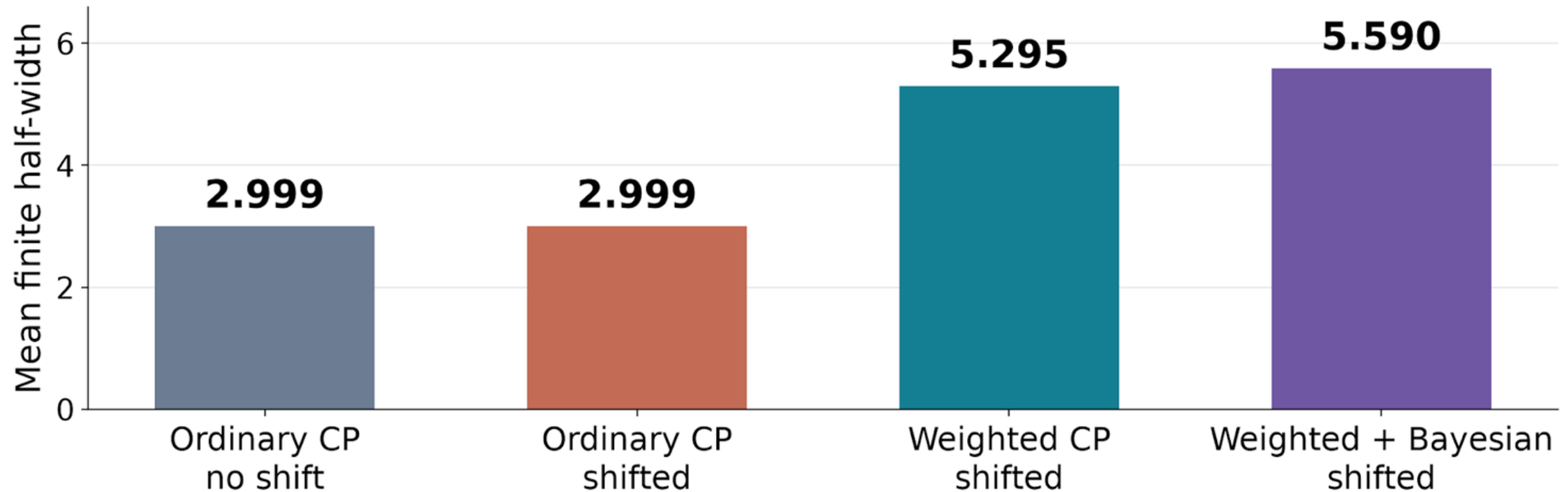
Coverage = fraction of test targets inside the band; goal = 90%



Weighted CP covers 1,814/2,000 targets, versus 1,221/2,000 without weighting.

Coverage recovery requires wider error bands

Mean interval half-width; smaller is useful only if coverage is adequate



The mean half-width grows from 3.00 to 5.29; the Bayesian maximum raises it to 5.59.

The arithmetic explains the coverage change

A half-width of 3 covers all easy errors and half the hard errors. In the original mixture, coverage is $0.8 \times 1 + 0.2 \times (3/6) = 0.90$.

After the shift, the same band covers only $0.2 \times 1 + 0.8 \times (3/6) = 0.60$. The predictor did not change; the tested population did.

To cover 90% of the new population, solve $0.2 + 0.8 \times (R/6) = 0.90$. This gives $R = 5.25$, close to the observed weighted mean half-width of 5.29.

Weighting changes how much each observed error contributes to the coverage threshold.

Next, validate coverage before changing learning

First, freeze a checkpoint and define which state–action population we want to cover. Use fresh calibration and test samples with defensible source-to-target ratios.

Then, integrate the query-dependent weighted threshold, preserve both BCA components, and verify weights, action alignment, checkpoint restoration, and infinite-band handling.

Finally, compare standard BCA with corrected weighted-threshold BCA across the declared seeds. Keep scale fitting and host losses fixed initially so the comparison isolates the correction.

Measure coverage and band width first; then ask whether harm ranking and regret improve.

What the next results will tell us

If weighted CP reaches the target coverage with useful band widths, the distribution-shift correction is working in that tested setting. It still needs a separate decision benefit.

If coverage improves but regret and harm ranking do not, we should examine the link between Bellman-target uncertainty and the host's action penalty.

If coverage remains poor or bands often become infinite, we should investigate the ratio, data overlap, sampling assumptions, and score construction before further training.

Success requires a clear chain: reliable bands → useful warnings → better decisions.

Warning quality changes with the situation

Collected states	Eligible / 64	Width AUROC	Distance AUROC
Host · reset	23	0.449	0.501
Host · step 100	34	0.640	0.652
BCA · reset	25	0.557	0.588
BCA · step 100	29	0.471	0.522

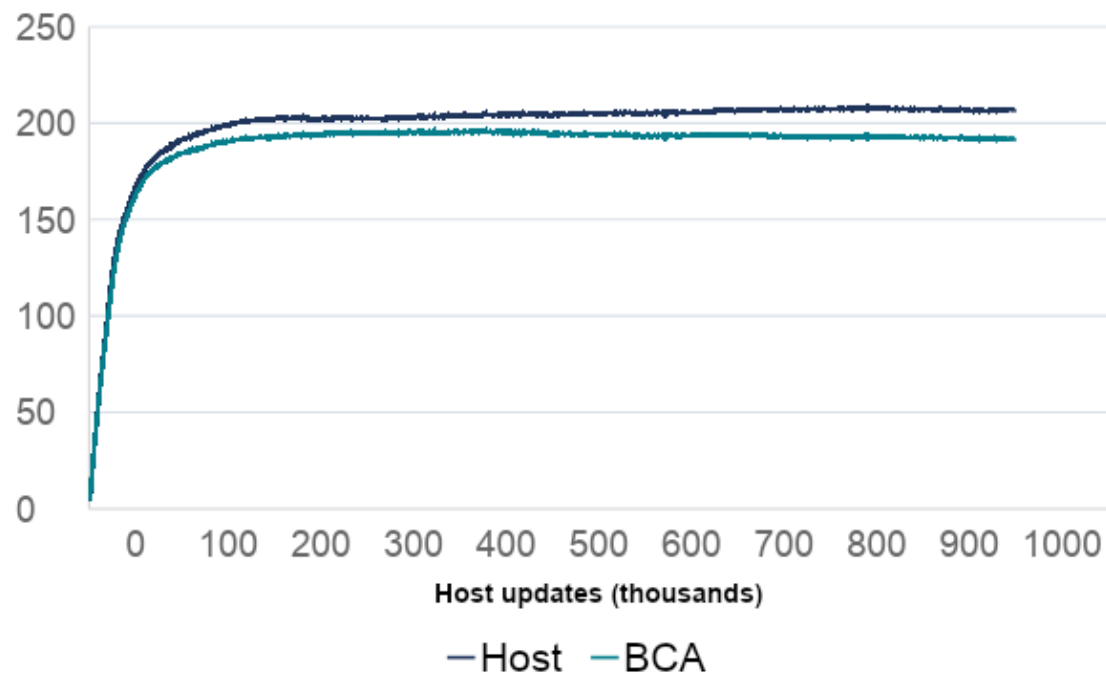
Widths vary: 1.35–2.48. No panel has all alternative widths tied.

Changing one global radius cannot repair which action is ranked as riskier.

TD3+BC: inspect Q and Bellman loss

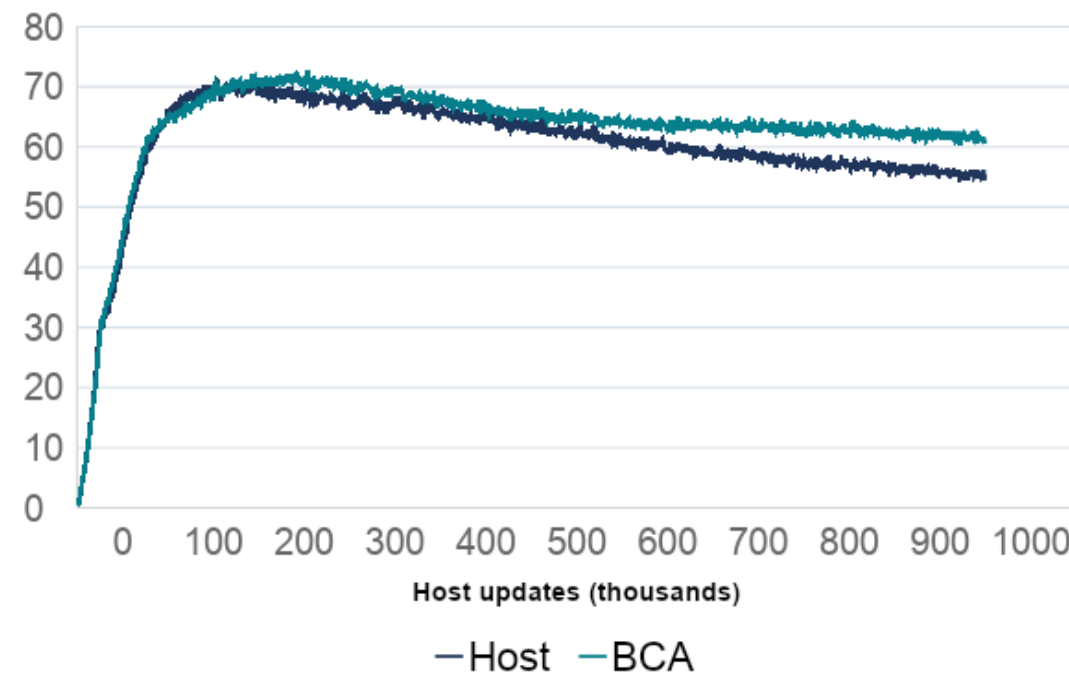
Q1 at proposed actor actions

Block mean Q estimate



Critic fitting loss

Block mean host loss



Final block Q: 207.10 → 191.92; critic loss: 55.42 → 61.31.

Lower Q does not establish more accurate Q. Larger loss does not identify OOD harm.

One reviewed Walker training seed. Training losses are fitting diagnostics, not labels for unsafe actions.

TD3+BC: lower Q does not establish better Q

On Walker, BCA lowers the final-block mean Q estimate from 207.10 to 191.92. This is consistent with more conservative values, but their accuracy is still unknown.

The critic fitting loss rises from 55.42 to 61.31. That loss compares predictions with a moving Bellman target; it does not measure error against known true returns.

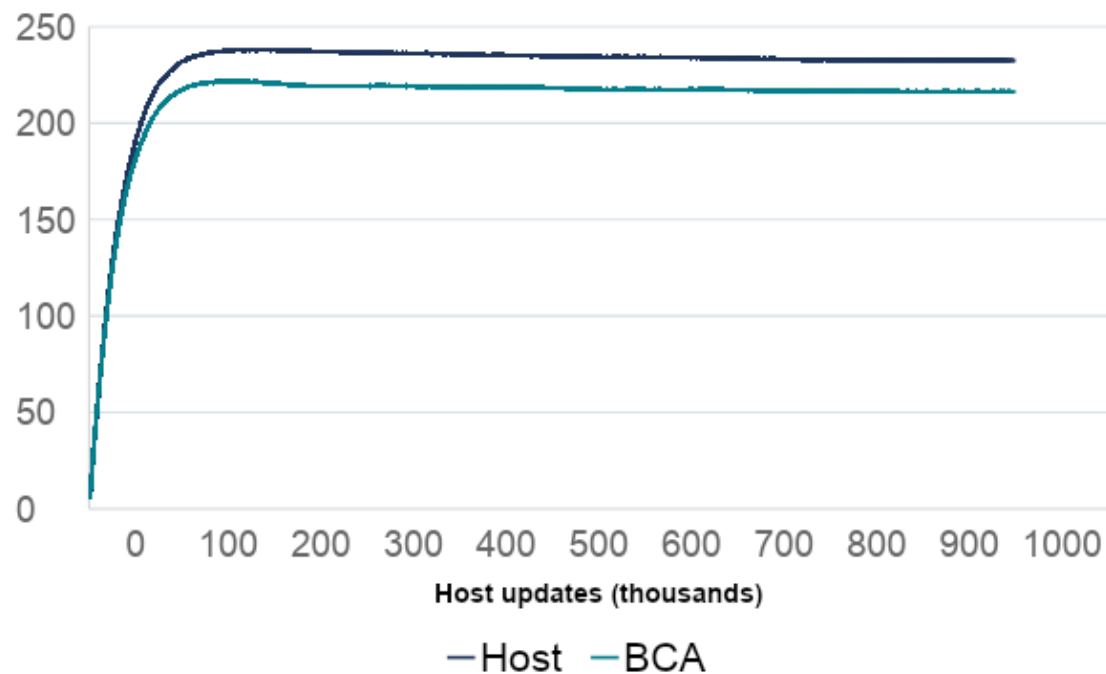
We need frozen-target coverage and paired action outcomes to decide whether the change is helpful. A lower Q estimate alone cannot explain improved or worse decisions.

Interpret these as training diagnostics until an independent outcome comparison supports a benefit.

ReBRAC: inspect Q and Bellman loss

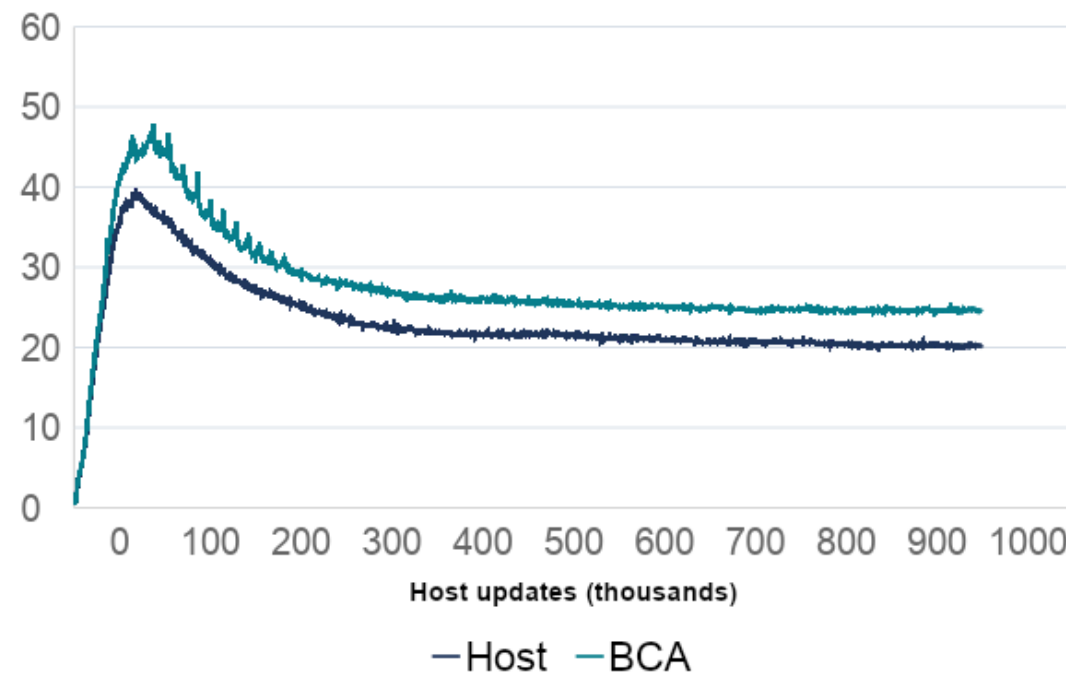
Minimum twin Q at recorded actions

Block mean Q estimate



Critic fitting loss

Block mean host loss



Final block Q: 232.63 → 216.31; critic loss: 20.30 → 24.75.

Lower Q does not establish more accurate Q. Larger loss does not identify OOD harm.

One reviewed Walker training seed. Training losses are fitting diagnostics, not labels for unsafe actions.

ReBRAC: the value tradeoff is also unresolved

On Walker, BCA lowers the final-block mean minimum Q from 232.63 to 216.31, while critic loss rises from 20.30 to 24.75.

BCA increases the actor's behavior-cloning strength, while the critic's behavior-cloning coefficient stays fixed. The change can influence later targets and values indirectly.

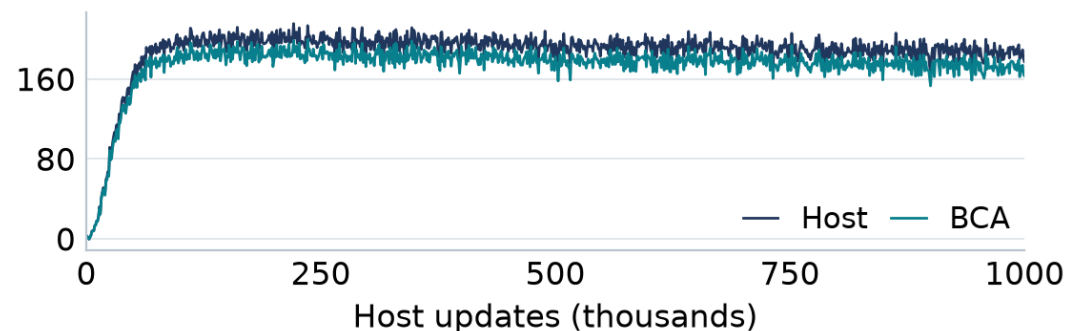
To assess the impact, compare fresh coverage and action outcomes at matched states. These logged losses cannot show that BCA corrected overestimation or detected OOD harm.

The mechanism is active; its effect on value accuracy and decisions still needs direct evidence.

CQL: inspect its critic and consumed penalty

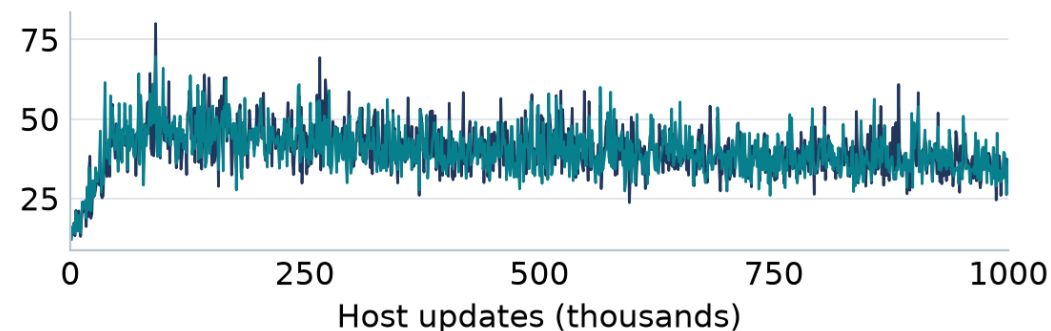
Q1 at recorded actions

Logged Q1



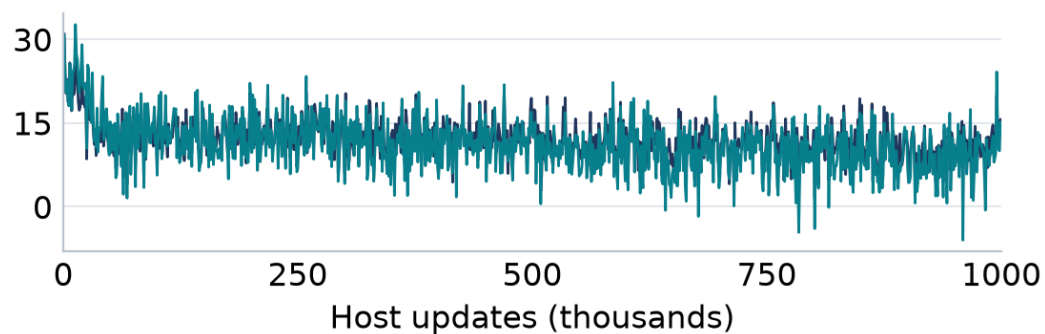
Bellman fitting loss

Logged objective



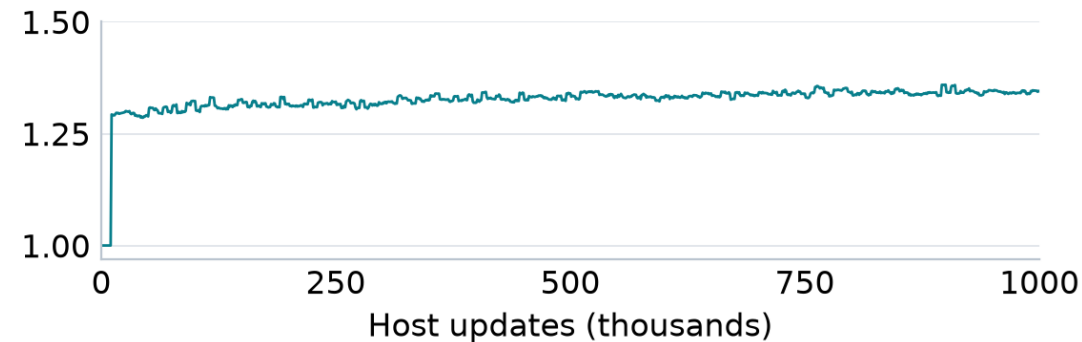
Conservative objective

Logged objective



Conservative-gap multiplier

Logged BCA multiplier



CQL's early diagnostics need a complete review

In the final retained Walker snapshot, BCA lowers Q1 from 177.15 to 163.12 and increases Bellman loss from 35.60 to 37.20.

The conservative objective changes from 13.68 to 13.21 with a BCA multiplier of 1.345. One recorded-action width scales the transition's conservative gap.

These are sparse snapshots, and independent checkpoint review remains pending. We must verify the full run and paired outcomes before attributing a benefit or failure to BCA.

Objective values show what the learner optimized; they do not directly measure action safety.

Fresh coverage is the missing real-data measurement

Coverage asks how often a new stated target lies inside its prediction band. The current BCA target is a one-step Bellman response, not known true long-run Q .

We have not verified fresh OOD response coverage for these real-RL results. Successful scale fitting, wide bands, and completed radius refreshes do not supply that measurement.

Freeze the model, use a fresh calibration/test design, and report coverage by the declared population alongside width and unsupported queries. Keep trajectory dependence explicit.

A nominal 90% setting states an objective; measured coverage establishes what actually happened.

Weighted CP changes how errors set the band

Step	Calculation	Purpose
Freeze the score	$S = \text{abs}(Y - m(X)) / \text{sigma}(X)$ $X = (\text{state}, \text{action})$	Compare target errors on a common positive scale.
Weight calibration errors	$w(X) = \text{target density} / \text{source density}$	Give more mass to errors more relevant to target queries.
Include the query	Threshold = $0.9 \times (\text{sum of bank weights} + w(\text{query}))$	Account for the unseen query in finite samples.
Find the conformal radius	First sorted error reaching the cumulative mass threshold	If no error reaches it, the radius is infinity.
Retain both components	$R = \max(R_{\text{CP}}(\text{query}), R_{\text{Bayes}})$ Half-width = $\text{sigma}(\text{query}) \times R$	Keep the conformal floor and Bayesian component visible.

Weighted CP needs a defensible target population

The target must have support in the source, and the conditional response law must stay the same. If state visitation changes, an action-only ratio is insufficient.

Deterministic actor outputs can be point masses relative to continuous source actions. A stochastic neighborhood would be a different target and must be declared explicitly.

Use fresh data after freezing the score, validate estimated ratios separately, and retain infinite intervals. Weight clipping and repeated bank reuse require additional justification.

The coverage claim concerns the declared response population; it is not a guarantee for every action.

Each host keeps its original BCA attachment initially

Host	Existing place BCA enters	What the integration must preserve
TD3+BC	Detached multiplier on actor behavior cloning	Delayed actor schedule and original host target.
ReBRAC	Detached multiplier on actor behavior cloning	Critic BC and regularized target construction.
CQL	Recorded-transition multiplier on the conservative gap	Original dual gap and actual query point for the band.
IQL	Shrinks capped AWR-weight excess above one	Shared Q/V, fixed gain 1 and original sampling semantics.
All four	Replace scalar-only radius with a query-aware calculation	Versioned checkpoints, finite/infinite behavior and parity.

The first correction isolates calibration. A different host penalty is a separate experiment.